

Article

Assessing the Social Factors Affecting Solar Energy Transition in the Mining Sector

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Abstract

This study examines the social factors shaping solar energy transitioning in Zambia's mining sector, a key contributor to the national resource economy. It focuses on how policies, incentive systems, communication channels, training and skills development, and stakeholder engagement and partnership influence the sector's readiness to adopt solar energy and support more reliable operations. A quantitative design was used, with a 5-point Likert-scale questionnaire administered to 192 respondents from mining companies, regulatory bodies, energy suppliers and local authorities. Data were analysed using descriptive statistics, reliability tests and hierarchical regression. The results show that stakeholder engagement and partnership and incentive systems are the strongest predictors of transition intentions, underscoring their role in improving energy security and supporting sustainable production. Policies, communication and training had weaker effects, indicating the need for stronger institutional coordination and targeted capacity building. Overall, the findings illustrate how social conditions shape renewable energy adoption and contribute to more resilient mining activities.

Keywords: social factors; solar energy; energy transition; mining sector; Zambia

1. Introduction

Meeting rising global energy demand while reducing dependence on unsustainable energy sources remains a central challenge of the twenty-first century. In response, countries are increasingly encouraged to diversify their energy mixes, strengthen the use of local renewable resources and adopt innovative practices that enhance energy security [1]. This transition is reinforced by Sustainable Development Goal 7, which promotes universal access to affordable, reliable and clean energy through renewable technologies such as solar, wind and thermal energy [1,2]. Achieving this goal is particularly critical for developing countries such as Zambia, where energy poverty, climate vulnerability and constrained infrastructure pose persistent development risks [3–5].

Zambia's energy sector reflects these pressures acutely [6–9]. Rapid population growth, projected to reach 26.9 million by 2035, is expected to intensify electricity demand across all sectors [7,10]. At the same time, the country's heavy reliance on hydropower exposes the system to increasing risks associated with drought, climate variability, ageing infrastructure



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and maintenance backlogs [11]. Within this context, solar energy offers a viable alternative due to Zambia's high solar potential, modular scalability and declining technology costs [12,13]. However, evidence from energy transition studies consistently shows that technological availability alone is insufficient to drive adoption. Social and institutional conditions, including policy support, incentive structures, institutional coordination and stakeholder acceptance, play a decisive role in shaping transition outcomes [5,14–16].

These dynamics are particularly pronounced in Zambia's mining sector, where renewable energy uptake remains limited despite growing energy constraints [17–20]. As the country's largest electricity consumer and a central driver of economic growth, the mining industry plays a critical role in shaping national energy demand and sustainability outcomes [21–23]. Reliable and affordable electricity supply is essential for continuous mineral production, cost control and operational stability. Recurrent power shortages and hydropower disruptions increase production risks and undermine sector performance. Solar energy presents an opportunity to improve energy reliability and enhance the resilience of mining operations; however, progress depends heavily on social conditions such as clear policy signals, accessible incentives, effective communication mechanisms and coordinated stakeholder engagement. Understanding how these social factors influence organisational decision-making is therefore essential for aligning industrial energy practices with national transition objectives and strengthening the long-term sustainability of the mining sector.

Against this background, this study investigates the social factors affecting solar energy transitioning in Zambia's mining sector. Specifically, it examines the relative influence of policies, incentive systems, communication channels, training and skills development, and stakeholder engagement and partnerships on organisational intentions to adopt solar energy within an energy-intensive industrial context. By focusing on these factors, the study responds to calls for greater empirical attention to the social dimensions of energy transition in emerging economies and resource-dependent sectors.

This study makes several original contributions to the renewable energy transition literature. First, it provides sector-specific empirical evidence from Zambia's mining industry, an energy-intensive sector that remains under-represented in transition studies despite its strategic importance. Second, it moves beyond conceptual discussion by quantitatively assessing and comparing the explanatory power of multiple social factors influencing solar transition intentions. Third, socio-technical transition theory is empirically operationalised by translating the social subsystem into measurable structural and people-centred variables and testing their relationships using hierarchical regression. Finally, the study contributes insights from an emerging economy context, addressing a literature gap that is predominantly dominated by evidence from developed energy markets.

2. Literature Review

Renewable energy systems based on locally available resources can enhance both environmental sustainability and economic resilience [24]. In Zambia, high solar potential presents a clear opportunity to strengthen energy security and reduce reliance on hydropower and imported fuels through decentralised generation systems [11]. However, evidence from energy transition research shows that the successful deployment of renewable technologies depends not only on resource availability but also on the social and institutional conditions that shape adoption decisions [5,16].

Recent studies emphasise that sustainability outcomes are strongly influenced by institutional coordination, governance structures and supporting infrastructure, alongside technological capability [25]. These factors are particularly salient in energy-intensive sectors such as mining, where operational continuity, investment risk and system reliability condition the feasibility of integrating solar energy into existing production systems. In

such contexts, social mechanisms influence transition pathways by shaping regulatory certainty, financial viability, organisational capacity and stakeholder alignment.

Guided by socio-technical transition theory [26], this study conceptualises solar energy transition as the outcome of interactions between social, institutional and technical systems rather than purely technological change. Within this framework, transition outcomes are shaped by governance structures, actor behaviour, institutional coordination and learning processes alongside technological capabilities. Accordingly, the study focuses on the social subsystem of the mining energy system and operationalises two core components: a structural system, comprising policies, incentive systems and communication channels, and a people-centred system, comprising training and skills development, and stakeholder engagement and partnerships. Together, these five social factors capture the key institutional, behavioural and coordination mechanisms through which renewable energy transition intentions are enabled or constrained in energy-intensive sectors such as mining. To ensure analytical relevance, the review prioritises recent literature on renewable energy transitions in emerging-market and industrial contexts, supplemented by foundational socio-technical studies that underpin the study's theoretical framing.

2.1. Policies

Energy policy influences renewable energy transitions by shaping regulatory certainty, investment expectations and institutional coordination [27]. Evidence from developed contexts shows that renewable uptake is strongest where long-term policy objectives are aligned with clear implementation responsibilities, as illustrated by Ireland's National Policy Statement on the Bioeconomy [24]. In Zambia, policies articulated under the National Energy Policy (2019) and the Eighth National Development Plan (2022–2026) express similar ambitions but remain weakly translated into enforceable and sector-specific implementation mechanisms [28,29]. This gap between policy intent and operational clarity weakens the credibility of regulatory signals required for long-term energy investment in the mining sector.

Zambia's renewable energy framework is underpinned by several legal and strategic instruments, including the National Energy Policy (2019), Vision 2030, the Electricity Act No. 11 of 2019, the Energy Regulation Act No. 12 of 2019 and the Renewable Energy Feed-in Tariff (REFiT) introduced in 2017 [5,30,31]. While these instruments promote diversification through solar, wind and biomass, empirical studies consistently find them insufficient to stimulate large-scale industrial adoption [5,32–35]. Although the NEP targets renewables supplying 40 per cent of total energy by 2030 [28], it lacks clear timelines, enforcement mechanisms and mining-specific provisions addressing firm power supply, energy storage and long-term power-purchase arrangements.

International experience shows that binding policy instruments, such as renewable portfolio standards and enforceable compliance targets, reduce uncertainty and accelerate adoption through predictable regulatory incentives [36,37]. By contrast, Zambia's framework relies largely on aspirational targets and administrative guidance, limiting its influence on firm-level behaviour. Despite recent efforts to strengthen licensing and enforcement [17,23,38], coordination gaps and limited support for system integration persist. Overall, Zambia's policy framework provides an important foundation for renewable energy development but offers weak behavioural signals for mining firms considering large-scale solar investment.

2.2. Incentive Systems

Financial incentives strongly influence renewable energy adoption by improving project financial viability and reducing perceived investment risk [39,40]. This pattern is

evident in Zambia's mining sector, where access to tax reliefs, concessional financing and other structured incentives remains a key driver of solar adoption. This highlights that economic feasibility, rather than awareness alone, shapes the pace of renewable transition in developing contexts. Instruments such as feed-in tariffs and net metering provide predictable revenue streams, shorten payback periods and reduce investment uncertainty, contributing to accelerated deployment in countries such as Germany, Spain and the UK [5,41].

In Zambia, the Renewable Energy Feed-in Tariff (REFiT) introduced in 2017 offers guaranteed tariffs for projects up to 20 MW [31,42], supported by tax exemptions, zero import duties on renewable equipment and recent green-budget measures, including green bonds and electric vehicle incentives [43]. Programmes such as Scaling Solar and the Rural Electrification Fund have expanded both utility-scale and off-grid renewable projects [44,45], with support from international partners including the African Development Bank and the European Union [5]. More recent initiatives, such as green bond guidelines and the Green Compact with the UK, further signal growing institutional support for renewable investment [17,46].

Despite these measures, most incentives remain oriented towards small-scale or rural applications rather than large industrial users. The absence of long-term, sector-specific financial instruments limits mining firms' ability to invest in stable, large-scale renewable systems capable of reducing exposure to power disruptions [5]. Expanding predictable and industry-tailored incentive schemes would therefore reduce investment uncertainty and support more consistent operational performance. As noted by Uhunamure et al. [39], well-designed financial incentives act as enabling mechanisms that align economic feasibility with renewable transition objectives in energy-intensive sectors such as mining.

2.3. Communication Channels

Effective communication channels influence solar energy transition by reducing information asymmetries and improving confidence among industry and community actors. Higher levels of knowledge and awareness have been shown to increase willingness to pay for renewable energy and strengthen adoption intentions [40]. This highlights the importance of clear, timely information flows within Zambia's mining sector, where targeted communication and training efforts can improve organisational readiness and acceptance of solar technologies. In practice, weak communication between policymakers, implementers and end-users has slowed renewable uptake in Zambia by limiting understanding of policy instruments, technical options and associated benefits, thereby reducing confidence in new technologies [47].

Within the mining sector, reliable data exchange systems further support coordination and operational security. Secure communication protocols help maintain the integrity of energy information networks and protect against unauthorised access [48], which is particularly important for energy planning in capital-intensive operations. At the community level, communication channels also play a critical role in maintaining social licence by clarifying project benefits, addressing concerns and managing expectations. Engagement approaches that use local languages, trusted leaders and appropriate media platforms can foster ownership and acceptance [5]. Without such mechanisms, renewable energy initiatives risk misinformation, resistance and limited community support, particularly in areas closely linked to mining activities.

2.4. Training and Skills Development

A major constraint to renewable energy adoption is limited training and skills development, which affects solar energy transition by strengthening or constraining organisational

and technical capacity to install, operate and maintain modern energy systems. Zambia faces notable skills shortages in renewable and energy-efficient technologies, restricting the mining sector's ability to transition effectively [49,50]. Bridging these gaps is essential for integrating renewable sources such as solar, wind, geothermal and biomass, which are projected to supply about 15 per cent of national energy use by 2030 [48]. Strengthening skills not only supports energy transition but also contributes to economic growth, employment creation and enterprise development along renewable value chains [51].

Training and skills development is particularly important in early-stage markets where technical expertise and operational experience remain limited [5]. The National Energy Policy (2019) identifies institutional capacity enhancement as a priority and calls for increased investment in research and innovation [28]. However, Zambia continues to experience shortages of specialised professionals and practical renewable energy expertise [5,52]. While development partners such as UNDP have introduced training and advisory support through Global Environment Facility initiatives, these programmes remain modest in scale [53]. Sustainable outcomes in the mining sector therefore depend on targeted capacity-building mechanisms that enable firms to internalise technical knowledge, reduce operational risk and sustain solar energy systems over the long term.

2.5. Stakeholder Engagement and Partnerships

Stakeholder engagement influences solar energy transition by enabling institutional coordination, reducing implementation conflict and maintaining social licence for renewable energy projects. The National Energy Policy (2019) highlights coordinated stakeholder participation as a key requirement for effective renewable energy implementation [28]. Talwar et al. [24] show that cooperative frameworks can significantly expand resource availability, as demonstrated in Ireland where collaboration between farms, industries and local authorities increased biomass supply. This insight is directly relevant to Zambia's mining sector, where partnerships between government agencies, mining companies and surrounding communities can provide the coordination and legitimacy needed for successful solar adoption.

In practice, weak engagement has slowed progress. Zambia's energy governance remains highly centralised, limiting local participation and constraining decentralised solutions [20]. Cases such as Mokambo peri-urban in Mufulira illustrate how limited consultation and poor awareness can delay renewable energy uptake [34]. Collaboration gaps among implementers, researchers and funding agencies further restrict progress [5,34]. Effective engagement therefore requires structured outreach mechanisms that align government, industry and community actors around shared objectives. Public-private collaboration aligned with national transition policies can strengthen implementation [17,54], although challenges persist due to fragmented planning, uneven technical capacity and weak accountability mechanisms [55]. Regional experience suggests that multi-stakeholder platforms linking government, financiers and civil society can improve inclusiveness and coordination. In Zambia, such efforts depend on supportive regulatory guidance and clearly defined governance responsibilities [56].

3. Materials and Methods

Guided by socio-technical transition theory [26], this study adopted a quantitative research design to examine how social subsystems influence organisational intentions to transition toward solar energy in Zambia's mining sector. The analytical framework focused on the social subsystem of the mining-energy system, distinguishing between structural factors (policies, incentive systems and communication channels) and people-centred factors (training and skills development, and stakeholder engagement and partnerships).

A questionnaire-based survey was used to collect structured data that allowed systematic assessment of stakeholder perceptions across these dimensions.

3.1. Questionnaire Design and Measurement

The research instrument was a closed-ended questionnaire developed from established empirical studies on renewable energy adoption and socio-technical transitions, and adapted to reflect the institutional and operational context of Zambia's mining sector [57]. The questionnaire comprised two sections: respondent demographics and measurement items for the key social variables. Each construct was measured using multiple items to capture its underlying dimensions: policies (5 items), incentive systems (5 items), communication channels (4 items), training and skills development (4 items), stakeholder engagement and partnerships (5 items), and transition intentions as the dependent variable (4 items). Item wording was contextualised to reflect mining-sector decision-making, regulatory exposure and energy reliability concerns.

Content validity was ensured through expert review by academics in energy policy and construction management, as well as practitioners with experience in Zambia's energy and mining sectors. Minor revisions were made to improve clarity and contextual alignment. Responses were recorded using a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). Full construct definitions and item wording are provided in Supplementary Table S1 to enhance transparency and reproducibility.

3.2. Sampling and Data Collection

Respondents were purposively sampled to target professionals in technical, managerial and strategic roles across mining companies, energy providers and regulatory institutions. Snowball sampling was subsequently used to identify additional knowledgeable participants [58]. Data were collected over a four-month period using an online Google Form (Google LLC, Mountain View, CA, USA) distributed via email invitations and follow-up reminders. In total, 192 valid responses were obtained.

The sample size was considered appropriate for capturing the diversity of Zambia's mining-energy stakeholder landscape, which is relatively concentrated and institutionally bounded. Zambia's mining sector comprises a limited number of large-scale mining firms, a small group of energy suppliers and a defined set of regulatory bodies rather than a large population of homogeneous actors. The final sample reflects representation across organisational types, ownership structures and experience levels, with respondents drawn primarily from the Copperbelt and North-Western regions, where mining activity and energy demand are most concentrated. The objective was therefore analytical representativeness rather than statistical generalisation, ensuring inclusion of informed actors across the mining energy system.

3.3. Data Analysis, Validity and Reliability Assessment

Descriptive statistics were computed to summarise central tendency and dispersion across variables. All statistical analyses were conducted using IBM SPSS Statistics for Windows, Version 27.0 (IBM Corp., Armonk, NY, USA). Internal consistency reliability was assessed using Cronbach's alpha, with values of 0.60 and above accepted given the exploratory nature of the study. Construct validity was evaluated using Exploratory Factor Analysis (EFA), supported by the Kaiser–Meyer–Olkin (KMO) measure and Bartlett's Test of Sphericity, which confirmed the suitability of the data for factor analysis.

Hierarchical regression analysis was employed to examine the incremental explanatory power of social factors on transition intentions. Variables were entered in a theory-driven sequence informed by socio-technical theory. Control variables were entered first, followed

by structural system variables, and finally people-centred variables, allowing assessment of the additional variance explained at each stage.

To improve transparency, the hierarchical regression model is specified as follows. Let TI_i denote the transition intention score for respondent i . The baseline model is

$$TI_i = \beta_0 + \beta_1 Size_i + \beta_2 Employees_i + \beta_3 Turnover_i + \beta_4 Tenure_i + \varepsilon_i \quad (1)$$

The full model is specified as:

$$TI_i = \beta_0 + \beta_1 Size_i + \beta_2 Employees_i + \beta_3 Turnover_i + \beta_4 Tenure_i + \beta_5 Policies_i + \beta_6 Incentives_i + \beta_7 Communication_i + \beta_8 Training_i + \beta_9 StakeholderEngagement_i + \varepsilon_i \quad (2)$$

where

β_0 is the intercept.

β_k are regression coefficients.

ε_i : The error term.

3.4. Control Variables

Control variables were selected based on prior literature on energy transition and organisational innovation, which identifies firm size, financial capacity and organisational maturity as background factors influencing technology adoption. Enterprise size, number of employees, annual turnover and organisational tenure were therefore included to account for baseline differences in capacity and structural positioning. These variables are commonly used in innovation and energy transition studies and allowed the analysis to isolate the explanatory power of social and institutional factors beyond differences in organisational scale and maturity. Other potential controls, such as corporate carbon footprint, were excluded due to limited availability of consistent emissions data and the perception-based nature of the survey instrument.

4. Results

This section presents the findings from the quantitative analysis of data collected from mining sector respondents on the social factors influencing solar energy transition (SET) in Zambia. The analysis integrates results from descriptive statistics, factor loading and reliability tests, and hierarchical regression models. These analyses provide an understanding of how policies, incentives, communication, training, and stakeholder engagement influence renewable energy readiness within the mining industry.

4.1. Demographic Descriptive Statistics

The following section presents the demographic characteristics of the respondents, providing context for interpreting the study's findings.

Table 1 summarises the demographic characteristics of 192 respondents representing mining companies, energy firms, and other related stakeholders. Engineers made up the largest group (38.5%), followed by managers (18.8%) and executive directors (7.3%), showing that most respondents were involved in technical and operational roles related to energy and production. Approximately 35.4% of participants had worked in their organisations for over ten years, indicating strong institutional experience. In terms of ownership, 33.3% of respondents were from state-owned institutions, 31.3% from privately owned entities, 13.5% from publicly listed firms, and 13.5% from joint ventures. Mining companies accounted for 43.8% of the sample, followed by government bodies (27.1%), energy companies (14.6%), and NGOs (6.3%). The composition of respondents across different

roles, ownership structures, and institutional types reflects a balanced representation of stakeholders within Zambia’s mining and energy landscape.

Table 1. Summary of Demographic Characteristics across Stakeholder Groups.

Category	Frequency	Percent	Mining Companies (<i>n</i> = 80)	Energy Companies (<i>n</i> = 28)	Other Mining Stakeholders (<i>n</i> = 84)
Respondent Role					
Executive Director	14	7.3%	5.0%	21.4%	8.3%
Managers	36	18.8%	18.8%	17.9%	20.2%
Engineer	74	38.5%	40%	39.3%	36.9%
Researcher	10	5.2%	0.0%	0.0%	11.9%
Other	58	30.2%	36.2%	21.4%	22.7%
Years in Organisation					
<1 years	14	7.3%	5.0%	14.3%	7.1%
1–5 years	58	30.2%	35.0%	35.7%	23.8%
6–10 years	52	27.1%	25.0%	25.0%	29.8%
>10 years	68	35.4%	35.0%	25.0%	39.3%
Organisation Ownership					
State-owned	64	33.3%	6.3%	28.6%	54.8%
Privately owned	60	31.3%	63.8%	28.6%	11.6%
Publicly listed	26	13.5%	20.0%	32.1%	9.5%
Joint venture/partnership	26	13.5%	7.5%	7.1%	14.3%
Other	16	8.3%	2.5%	3.6%	9.5%
Type of Organisation					
Mining Company	84	43.8%	100%	0.0%	0.0%
Government Body	52	27.1%	0.0%	0.0%	61.9%
Energy Company	28	14.6%	0.0%	100.0%	9.5%
NGOs	12	6.3%	0.0%	0.0%	14.3%
Association	8	4.2%	0.0%	0.0%	9.5%
Other	8	4.2%	0.0%	0.0%	4.8%

4.2. Kruskal–Wallis

The results of the Kruskal–Wallis test are summarised in Table 2. A Kruskal–Wallis test was conducted to examine whether future intentions to transition differed across three respondent groups: mining companies, energy suppliers and other mining stakeholders. This nonparametric test provides a suitable alternative to one-way ANOVA when comparing three or more independent groups [59]. The analysis used the intention scores as the dependent variable.

Table 2. Summary of Kruskal–Wallis Test Results for Group Differences.

Hypothesis Test Summary				
	Null Hypothesis	Test	Sig. ^{a,b}	Decision
1	The distribution of Scores is the same across categories of Groups.	Independent-Samples Kruskal–Wallis Test	0.873	Retain the null hypothesis.

^a. The significance level is 0.050. ^b. Asymptotic significance is displayed.

The hypothesis that ‘the distribution of scores is the same across categories of groups’ was statistically supported and thus, upheld. This was because the *p*-value of the data was greater than the significance level ($p > \alpha$; $0.873 > 0.05$). The implication of this result is that the scores on future intentions to transition by mining companies and scores by mining stakeholders were almost the same, the difference was not significant. The scores were relatively consistent across the three respondent groups, indicating similar views on energy

transitioning in the mining sector. As such, it is appropriate to combine the responses from all three groups for aggregate analysis.

4.3. Descriptive Statistics

Descriptive statistics were used to summarise the main patterns in the data and provide an overview of respondents' perceptions. Measures such as means, standard deviations, skewness and kurtosis helped show central tendencies, variability and the distribution shape. These results offered a basic understanding of the responses before moving to more advanced analyses.

Table 3 summarises construct-level descriptive statistics for the social variables influencing solar energy transition in Zambia's mining sector. Detailed item-level statistics are provided in Supplementary Table S2 to ensure transparency. Mean values across the five constructs range from approximately 2.96 to 3.28, indicating moderate agreement and emerging readiness for solar adoption.

Table 3. Descriptive Statistics for the Mining Sector.

Construct	Mean	Std. Deviation
	Statistic	Statistic
Policies	3.21	0.24
Incentive System	3.03	0.30
Communication Channels	2.96	0.21
Training and Skills Development	3.28	0.61
Stakeholder Engagement and Partnerships	3.22	0.48

Training and skills development recorded the highest mean (Mean = 3.28, SD = 0.61), highlighting recognition of capacity as a key enabler, although variability suggests uneven skills and coordination. Stakeholder engagement and partnerships also showed a positive mean (Mean = 3.22, SD = 0.48), underscoring the perceived importance of collaboration in energy-intensive sectors. Policy frameworks and incentive systems exhibited similar moderate means (Policies: Mean = 3.21, SD = 0.24; Incentives: Mean = 3.03, SD = 0.30), reflecting awareness of policy and financial instruments alongside concerns about effectiveness and access. Communication channels recorded the lowest mean (Mean = 2.96, SD = 0.21), pointing to persistent information and coordination gaps. Overall, the results indicate partial institutional readiness for solar energy transition, constrained by implementation and communication challenges. These descriptive patterns provide context for the subsequent regression analysis examining the relative influence of social factors on transition intentions.

4.4. Validity and Reliability

Given the exploratory nature of this study and the limited availability of context-specific measurement scales for solar energy transitioning in Zambia's mining sector, Exploratory Factor Analysis (EFA) was employed to assess construct validity [60]. EFA is well suited to emerging research contexts, particularly in developing-country settings, where underlying factor structures require empirical identification. Sampling adequacy was evaluated using the Kaiser–Meyer–Olkin (KMO) measure and Bartlett's Test of Sphericity, both of which confirmed the suitability of the data for factor analysis. Internal consistency reliability was assessed using Cronbach's alpha, with all constructs exceeding the accepted threshold of 0.70, indicating satisfactory reliability. These validity and reliability assessments were complemented by regression diagnostics to confirm that statistical assumptions

were met and that the empirical results were not driven by estimation bias. Collectively, these procedures strengthen the robustness and credibility of the findings in the absence of longitudinal data. The sampling adequacy and suitability of the data for factor analysis are assessed using the Kaiser–Meyer–Olkin and Bartlett’s tests, as summarised in Table 4.

Table 4. KMO and Bartlett’s Test Statistics for Factor Analysis is a table.

KMO and Bartlett’s Test		
Kaiser-Meyer-Olkin Measure of Sampling Adequacy		0.785
Bartlett’s Test of Sphericity	Approx. Chi-Square	1280.592
	df	231
	Sig.	0.000

The KMO value of 0.785 indicated that the sample size was adequate for factor analysis and exceeded the recommended minimum threshold of 0.50, demonstrating strong sampling adequacy [1,2]. This value also surpassed the 0.70 benchmark commonly associated with good construct validity, confirming that the correlation patterns among variables were appropriate for dimension reduction. Bartlett’s Test of Sphericity was statistically significant ($\chi^2 = 1280.592$, $df = 231$, $p < 0.001$), suggesting that the correlation matrix was not an identity matrix and that the variables shared sufficient common variance to justify factor extraction. Together, the KMO results, Bartlett’s test outcome, and internal reliability coefficients confirm that the questionnaire was both statistically valid and reliable, supporting its suitability for examining the social factors influencing solar energy transitioning in the mining sector.

4.5. Factor Loadings and Reliability Statistics

The table presents the final retained items, factor loadings, and reliability statistics for all five constructs measuring social factors influencing solar energy transitioning in the mining sector. Overall, the results show that the constructs demonstrate strong internal consistency and acceptable validity for exploratory factor analysis. The factor loadings and reliability statistics for the retained social subsystem constructs are presented in Table 5.

Table 5. Factor Loadings and Reliability Statistics for Social Subsystem Constructs.

Variable	Undeleted Items of Short Code Questionnaire Items Used to Operationalize	Factor Loadings	Reliability Stats. (Cronbach’s Alpha)
Future intention to transition	The mining sector is committed to integrating solar energy	0.688	$\alpha = 0.928$ (4 items)
	The mining sector increasingly prioritises solar energy adoption	0.804	
	The mining sector aims to reduce reliance on conventional energy through solar adoption.	0.781	
	Building internal capacity for solar transition is a strategic priority in mining.	0.771	

Table 5. *Cont.*

Variable	Undeleted Items of Short Code Questionnaire Items Used to Operationalize	Factor Loadings	Reliability Stats. (Cronbach's Alpha)
Policies	National energy policies effectively promote sustainable energy transitioning.	0.578	$\alpha = 0.881$ (4 items)
	National energy policies are adequate to support renewable energy adoption and integration.	0.673	
	National mining policies align with sustainable energy goals and environmental responsibilities.	0.674	
	National policies provide clear guidance and incentives for organisations to adopt sustainable energy practices	0.725	
Incentive System	Government-provided incentives are adequate	0.601	$\alpha = 0.770$ (3 items)
	Mining companies use government incentives to adopt sustainable energy.	0.719	
	The process of accessing government incentives for solar energy is straightforward for Mining companies.	0.746	
Communication Channels	The government communicates clear strategies for solar energy adoption.	0.768	$\alpha = 0.884$ (4 items)
	The government provides guidance on compliance with solar energy regulations.	0.810	
	The government provides updates on targets for renewable energy use.	0.776	
	Information on government incentives for solar energy is easily accessible.	0.763	
Skills Development	Industry leaders actively support initiatives to improve energy efficiency	0.618	$\alpha = 0.790$ (2 items)
	Roles and responsibilities for achieving sustainable energy goals are well-defined across the mining sector	0.741	
Stakeholder Engagement & Partnerships	Mines engage external stakeholders in solar initiatives.	0.829	$\alpha = 0.859$ (3 items)
	Mines have established partnerships to support solar energy projects.	0.856	
	Mines collaborate with local communities to address concerns and maximise the benefits of solar energy projects.	0.863	

The policy construct retained four items with factor loadings ranging from 0.578 to 0.725, indicating that each item contributes meaningfully to the underlying concept. These items capture perceptions of policy effectiveness, adequacy, alignment with sustainability

goals, and clarity in providing incentives. The construct has a Cronbach's Alpha of 0.881, which indicates excellent internal reliability. This means respondents consistently interpreted policy-related statements, confirming that the construct is stable and well-defined. Three items were retained for the incentive system construct, showing factor loadings between 0.601 and 0.746. These loadings demonstrate moderate to strong relationships between the items and the construct. The Cronbach's Alpha value of 0.770 reflects acceptable reliability, suggesting that respondents viewed incentive adequacy, ease of access, and utilisation in a generally consistent manner. Compared to the policy construct, the reliability is slightly lower, which suggests more variation in how respondents experienced or understood government incentives. The communication channels construct has four items, all showing strong factor loadings between 0.763 and 0.810. These values indicate a clear and coherent construct where each item strongly aligns with the underlying dimension. The reliability score of 0.884 reflects very high internal consistency. Overall, this suggests that respondents had similar views about government communication clarity, regulatory guidance, renewable energy updates, and accessibility of information. Two items were retained under training and skills development, with factor loadings of 0.618 and 0.741. Both loadings are acceptable for exploratory factor analysis, especially given the small number of items. The Cronbach's Alpha of 0.790 signifies good reliability, meaning that respondents consistently linked leadership support and the clarity of roles and responsibilities to training and skills development efforts in the sector. This construct therefore remains stable despite having only two indicators. The stakeholder engagement construct retained three items, with the only displayed loading (for the first item) being 0.829, which is very strong. Although the specific loadings for the remaining two items are not shown in the table, the Cronbach's Alpha of 0.859 indicates high internal reliability across all three indicators. This suggests that respondents consistently associated external engagement, partnerships, and community collaboration as core components of stakeholder engagement in solar energy transitioning.

Across all constructs, the factor loadings exceed the minimum acceptable threshold of 0.50, confirming that each retained item meaningfully contributes to the measurement of its respective variable. The reliability statistics, ranging from 0.770 to 0.884, further demonstrate that the constructs are internally consistent and suitable for continued analysis. These results confirm that the measurement instrument used in this study is both reliable and valid for examining social factors influencing solar energy transitioning in Zambia's mining sector.

4.6. Correlation Analysis

This section summarises the correlation results that explore how the key variables relate to future intentions to transition to renewable energy. Table 6 presents the Pearson correlation coefficients for the study variables, showing both the strength and direction of their associations. The results indicate which social subsystem factors are significantly linked with transition intentions and therefore provide an early indication of where influence on transition readiness is most pronounced within the sector.

Table 6 presents the correlation analysis between Future Intentions to Transition (FUINTR) and the social subsystem variables included in the study. These variables form part of the socio-technical framework that explains how organisational and institutional conditions shape solar energy transition readiness in the mining sector. The results show that stakeholder engagement has the strongest positive association with transition intentions ($r = 0.482$, $p < 0.01$), suggesting that organisations with stronger external partnerships and collaborative practices are more inclined to adopt renewable energy solutions. Policies also show a positive, though weaker, correlation with transition intentions ($r = 0.208$,

$p < 0.05$), indicating that clearer or more supportive policy signals may contribute to improved transition readiness. Incentive systems, communication channels and training and skills development show positive but non-significant correlations with FUINTR ($r = 0.195$, $p = 0.056$; $r = 0.140$, $p = 0.172$; and $r = 0.170$, $p = 0.098$, respectively). Although these variables do not reach statistical significance, their positive direction suggests that improvements in these areas may still contribute to transition readiness. Overall, the correlation results highlight stakeholder engagement as the most influential factor at this stage, reinforcing the importance of collaboration and interaction among actors within the mining energy ecosystem.

Table 6. Pearson Correlation Matrix for Transition Variables.

		FUINTR	POLI	INSYS	COMCHA	SKLDEV	STKENG
FUINTR	Correlation	1					
	Sig. (2-tailed)						
POLI	Correlation	0.208 *	1				
	Sig. (2-tailed)	0.042					
INSYS	Correlation	0.195	0.582 **	1			
	Sig. (2-tailed)	0.056	0.000				
COMCHA	Correlation	0.140	0.553 **	0.744 **	1		
	Sig. (2-tailed)	0.172	0.000	0.000			
SKLDEV	Correlation	0.170	0.435 **	0.501 **	0.652 **	1	
	Sig. (2-tailed)	0.098	0.000	0.000	0.000		
STKENP	Correlation	0.482 **	0.359 **	0.307 **	0.414 **	0.513 **	1
	Sig. (2-tailed)	0.000	0.000	0.002	0.000	0.000	
	Sig. (2-tailed)	0.177	0.017	0.045	0.551	0.560	0.931
	N	192	192	192	192	192	192

Note: $p < 0.05$ (*), $p < 0.01$ (**), two-tailed.

4.7. Regression Analysis

A multiple regression analysis was carried out to determine how social subsystem factors influence renewable energy transition readiness in Zambia's mining sector. Guided by the Socio-Technical Theory, the analysis focused on policies, incentive systems, communication channels, training and skills development and stakeholder engagement and partnerships to see how they contribute beyond basic organisational characteristics. Control variables were entered first, followed by the social predictors, which made it possible to assess the added explanatory value of each group. The results are presented through correlations, model summaries and regression coefficients and are discussed in relation to developments in the mining and energy sectors. The results of the analysis of variance (ANOVA) for the regression model are presented in Table 7.

Table 7. Analysis of Variance.

ANOVA ^a						
	Model	Sum of Squares	df	Mean Square	F	Sig.
1	Regression	43.280	14	3.091	5.728	0.000 ^b
	Residual	43.719	81	0.540		
	Total	86.999	95			

^a. Dependent Variable: FUINTR. ^b. Predictors: (Constant), POLI, INSYS, COMCHA, TSKLDEV, STKENG.

The ANOVA results assess whether the overall regression model significantly predicts the dependent variable FUINTR (presumably “Future Intentions for Solar Energy Transitioning”). The model includes five predictors: POLI, INSYS, COMCHA, TSKLDEV and STKENG. The regression sum of squares is 43.280 with 14 degrees of freedom, while the residual sum of squares is 43.719 with 81 degrees of freedom. This yields an F-value of 5.728, with a corresponding significance level of $p = 0.000$. This result indicates that the model is statistically significant at the 0.01 level, the social factors included in the model collectively provide a meaningful explanation of the variation in FUINTR. The significant F-value shows that the model performs better than one with no predictors, confirming that at least one of the independent variables makes a significant contribution to predicting future intentions for solar energy transitioning in the Zambian mining sector.

4.8. Hierarchical Regression

Hierarchical regression analysis was employed to assess how social subsystem variables influence intentions to transition to solar energy after accounting for basic organisational characteristics. The regression followed a theory-driven sequence informed by socio-technical theory. Model 1 included organisational control variables (firm size, employee numbers, turnover, and tenure) to account for baseline differences in organisational capacity. Model 2 introduced the structural and enabling social factors; policies, communication channels, and training and skills development which reflect the institutional and informational context shaping transition decisions. Model 3 subsequently added the more agency-driven social factors, namely incentive systems and stakeholder engagement and partnerships, to assess their incremental explanatory power beyond the weaker structural variables. The inclusion of the social subsystem variables increased the explained variance in transition intentions from $R^2 = 0.016$ to $R^2 = 0.201$ ($\Delta R^2 = 0.185$, $p < 0.01$). Notably, stakeholder engagement emerged as the strongest contributor, indicating that participatory and collaborative mechanisms exert a greater influence on transition intentions after controlling for organisational characteristics and broader structural conditions. This sequencing allowed for a clear assessment of how explanatory power improved at each stage through changes in R^2 and associated F-change values.

To assess the robustness of the empirical findings, several complementary diagnostic and validation procedures were applied. Multicollinearity was examined using variance inflation factors (VIF), with all values remaining below the conservative threshold of 3, indicating that intercorrelations among predictors did not distort coefficient estimates. Given the perceptual and cross-sectional nature of the survey data, common method bias was assessed using Harman’s single-factor test, which showed that no single factor accounted for the majority of variance, suggesting that common method effects were unlikely to drive the results. In addition, the stability of the regression estimates was examined through alternative model specifications, including reordering of predictor blocks and exclusion of non-significant variables, with the substantive pattern of results remaining unchanged. These checks increase confidence that the observed dominance of stakeholder engagement and incentive systems reflects genuine structural relationships rather than artefacts of model specification or measurement design. Table 8 presents the hierarchical regression results for the predictors of solar energy transition intentions, showing the incremental contribution of control variables and social subsystem factors across the three models.

The analysis began with a baseline model including only organisational control variables (tenure, number of employees, annual turnover and organisation size). Model 1 explained a negligible proportion of variance in solar energy transition intentions ($R^2 = 0.016$),

and none of the control variables were statistically significant, indicating that organisational demographics alone do not adequately explain variation in transition readiness.

Table 8. Hierarchical Regression Results for Predictors of Solar Energy Transition Intentions.

Variables	Model 1 (Controls Only)	Model 2 (+Social Subsystem Predictors)	Model 3 (Final Model)
Control Variables			
Tenure	$\beta = 0.022$	$\beta = -0.029$	$\beta = -0.029$
No. of employees	$\beta = -0.065$	$\beta = -0.027$	$\beta = -0.027$
Annual turnover	$\beta = -0.032$	$\beta = -0.042$	$\beta = -0.042$
Organisation size	$\beta = 0.101$	$\beta = 0.010$	$\beta = 0.010$
Independent variables			
POL	-	$\beta = -0.012$	$\beta = -0.012$
INSYS	-	$\beta = 0.271$	$\beta = 0.271$
COMCHA	-	$\beta = -0.166$	$\beta = -0.166$
TSKLDEV	-	$\beta = -0.035$	$\beta = -0.035$
STKENP	-	$\beta = 0.479$	$\beta = 0.479$
MODEL FIT			
R ²	0.016	0.201	0.201
ΔR^2	-	0.185	0.000
F	0.288	3.738 **	3.738 **

Notes: ** $p < 0.01$.

Model 2 introduced the five social subsystem predictors derived from socio-technical theory: policies, incentive systems, communication channels, training and skills development, and stakeholder engagement and partnerships. Entering this block of variables simultaneously resulted in a substantial improvement in model fit, increasing the explained variance to 20.1 per cent ($\Delta R^2 = 0.185$, $p < 0.01$). This ΔR^2 therefore reflects the combined explanatory contribution of the social subsystem predictors, rather than the effect of any single variable in isolation. Within this block, stakeholder engagement emerged as the strongest positive predictor of transition intentions ($\beta = 0.479$), followed by incentive systems ($\beta = 0.271$). In contrast, policies ($\beta = -0.012$), communication channels ($\beta = -0.166$) and training and skills development ($\beta = -0.035$) did not exhibit statistically significant effects. The relative magnitude of the standardised coefficients indicates that the observed increase in explained variance is driven primarily by stakeholder engagement and incentive systems.

Model 3 retained the same set of predictors and therefore produced identical results, confirming the stability of the findings. Overall, the limited explanatory role of organisational controls and the concentration of explanatory power within specific social constructs reinforce the view that solar energy transition intentions in Zambia's mining sector are shaped less by structural characteristics or policy presence alone and more by effective stakeholder coordination and credible financial incentives.

5. Discussion

The study examined the social factors influencing solar energy transition (SET) in Zambia's mining sector. Drawing on socio-technical theory, it explored how the interaction between social, institutional, and technical systems shapes the mining industry's readiness for renewable energy adoption. The discussion links the empirical results to the reviewed literature and interprets their meaning within the wider context of Zambia's energy transition efforts.

5.1. Policies-Related Findings

The findings show that although respondents acknowledged the existence of policies supporting renewable energy, these policies did not significantly predict solar adoption intentions. This aligns with observations by Mhango [5] and Chitandula et al. [11], who

argue that Zambia's policy environment is well articulated but weak in delivery. Instruments such as the National Energy Policy (2019), the Electricity Act No. 11 of 2019 and the Renewable Energy Feed-in Tariff (2017) signal government commitment, yet they lack the operational clarity required to drive meaningful uptake within the mining sector [28,31]. Pegels [32] similarly notes that policy frameworks in developing-country contexts often underperform when enforcement, accountability and resource allocation mechanisms are insufficient, while Wojnarowska et al. [25] emphasise the importance of evidence-based assessment in guiding effective sustainability transitions.

Beyond their descriptive intent, Zambia's national energy policies exhibit several structural contradictions that constrain effective implementation within the mining sector. While policy instruments promote renewable energy diversification, they are not consistently supported by sector-specific operational guidelines, coordinated institutional responsibilities or clearly defined enforcement mechanisms. This disconnect creates uncertainty for mining firms that require predictable regulatory signals to support long-term energy investment decisions. As a result, policies function more as aspirational frameworks than as actionable transition tools, limiting their influence on firm-level behaviour despite their stated objectives.

The weaker predictive effect of policy variables reflects a clear distinction between the existence of policy frameworks and their perceived effectiveness in practice. From a socio-technical perspective, policies influence transition behaviour primarily by shaping regulatory credibility, investment certainty and coordination across actors; where enforcement mechanisms, sector-specific guidance and predictable implementation pathways are weak, policy frameworks generate limited behavioural signals for firm-level decision-making. In the mining sector, where energy decisions are shaped by operational risk, cost certainty and supply reliability, policies that lack enforceable instruments, sector-specific guidance, predictable timelines and financial integration provide weak behavioural signals. Consequently, while Zambia's policy instruments establish a necessary foundation for renewable energy growth, they do not yet deliver the credibility or implementation pathways required to influence firm-level transition intentions or reduce energy-related operational vulnerabilities.

5.2. Incentive Systems-Related Findings

Incentive systems exhibited a positive and statistically significant influence on solar energy transition intentions, underscoring the central role of financial viability in shaping corporate energy decisions. This finding aligns with studies by Kılıç and Kekezoglu [41] and Chitandula et al. [17], which highlight the importance of financial instruments such as tax exemptions, feed-in tariffs and concessional loans in expanding renewable energy adoption. However, the relatively modest mean scores observed in this study suggest that access to such incentives in Zambia remains limited or administratively complex, particularly for large industrial users. Previous research similarly indicates that existing incentive structures are more oriented towards small-scale or off-grid applications than towards energy-intensive sectors such as mining [5,11].

While incentive-based renewable energy policies in countries such as Germany and Spain have been effective in accelerating solar deployment, their direct replication in Zambia is constrained by differences in institutional capacity, market structure and financial depth. European incentive regimes operate within stable regulatory environments, liberalised electricity markets, mature grid infrastructure and well-developed green finance ecosystems. In contrast, Zambia's energy sector is characterised by a vertically structured electricity market, limited fiscal space, evolving regulatory enforcement and constrained access to long-term finance. Consequently, the portability of international experience lies

less in replicating specific subsidy levels or tariff designs, and more in adapting core principles such as policy credibility, administrative simplicity and investor certainty to local institutional realities.

From an engineering and financial perspective, the relevance of incentive systems is further reinforced by global declines in photovoltaic technology costs and the emergence of green financing instruments. Over the past decade, falling module prices, improved system efficiencies and advances in energy storage have reduced the capital intensity of solar projects, making them increasingly viable for industrial applications. However, the ability of mining firms to capitalise on these trends depends on access to appropriate financial mechanisms, including concessional loans, blended finance, green bonds and structured power-purchase agreements. In Zambia, the limited penetration of such instruments constrains the translation of technological cost reductions into realised investment decisions, highlighting the importance of aligning financial innovation with institutional incentives and stakeholder coordination.

Insufficient or poorly targeted incentives limit the ability of mining companies to invest in stable and affordable energy systems, with direct implications for production reliability and operational costs. The findings therefore suggest that expanding incentive schemes tailored to industrial energy needs, alongside simplifying access procedures, would support wider renewable energy uptake. Well-designed financial instruments can act as enabling conditions that align social and technical subsystems, facilitate innovation diffusion across mining operations and strengthen long-term sector performance.

5.3. Communication Channels-Related Findings

Communication channels recorded low mean values and non-significant regression results, indicating continuing weaknesses in information flow between government, regulators and industry. Poor communication has previously been identified as a barrier to effective energy policy implementation in Zambia [16,47]. Although the construct showed high reliability, its weak statistical influence suggests that information-sharing mechanisms are present but not operating effectively. Limited awareness of incentives, unclear technical guidelines and irregular updates on national targets reduce confidence in the transition process and make coordinated planning more difficult.

Communication acts as a link between policy design and the practical uptake of new technologies. Strengthening this link through transparent data systems, structured consultations and consistent public reporting would support better decision-making, improve alignment across organisations and help reduce disruptions associated with energy uncertainty. Clearer and more efficient communication would therefore create conditions that encourage smoother adoption of solar energy within mining operations.

5.4. Training and Skills Development-Related Findings

Training and skills development showed no significant influence on solar energy transition in this study. Respondents noted that while some technical capability exists, structured training, coordinated learning and practical exposure remain limited. This aligns with earlier work highlighting gaps in human capital and institutional capacity as major barriers to renewable energy adoption in Zambia [2,5]. Limited access to regular training and skills development affects how confidently organisations can prepare for new technologies and increases uncertainty when planning transition activities.

Strengthening training systems would improve familiarity with solar technologies, enhance problem-solving capacity and support more informed decision-making within mining operations. Better-skilled personnel are also likely to manage energy systems more efficiently, reduce operational disruptions and contribute to more stable and reliable pro-

duction. Targeted up skilling, industry academic partnerships and continuous professional development would therefore help create conditions that support a smoother and more sustainable shift to solar energy.

5.5. Stakeholder Engagement and Partnerships-Related Findings

Stakeholder engagement and partnerships emerged as the strongest and most consistent predictor of solar energy transition intentions, underscoring the central role of coordination and collective action in enabling renewable energy adoption. This finding aligns with earlier studies showing that effective collaboration among government agencies, mining companies, financiers and communities is critical for overcoming institutional fragmentation and implementation barriers in energy transitions [5,11,22,27]. Through shared decision-making, information exchange and risk distribution, engagement mechanisms reduce uncertainty and strengthen confidence in new energy investments.

Evidence from South Africa [51] and Zambia further indicates that renewable energy uptake is shaped not only by technical capacity but also by social and behavioural conditions. Policy support, affordability and engagement interact to influence organisational readiness, particularly in energy-intensive sectors such as mining. As noted by Nyokabi et al. [53], cooperation among diverse actors enhances accountability, resource mobilisation and knowledge transfer, while partnerships with research institutions and technology providers facilitate solutions aligned with operational mining requirements [52]. In Zambia, weak engagement structures have limited project scalability and slowed implementation, reflecting gaps in coordination rather than a lack of technological potential.

The strong predictive effect of stakeholder engagement observed in this study reinforces the socio-technical perspective that energy transitions advance most effectively when actor networks are aligned around shared objectives and coordinated governance arrangements. By enabling trust, coordination and institutional learning, stakeholder partnerships function as a key transmission channel through which social conditions translate into concrete transition intentions within the mining sector.

5.6. Broader Implications

The findings indicate that progress toward renewable energy transition in Zambia's mining sector is shaped primarily by social conditions, which remain uneven across institutions. Policy frameworks alone are insufficient without credible incentives, effective communication and coordinated stakeholder engagement. Financial viability and collaborative arrangements exert greater influence on transition outcomes than broad regulatory statements or stand-alone training initiatives.

Although grounded in Zambia's mining sector, the findings are analytically transferable to other African mining contexts with similar energy profiles. Mining industries across the continent are characterised by high energy intensity, exposure to supply disruptions and capital-intensive production systems. In such settings, incentive credibility, institutional trust and stakeholder coordination are likely to play a decisive role in shaping renewable energy adoption, although the relative influence of policy variables may increase where regulatory enforcement is stronger.

The results also hold relevance for other energy-intensive industries in Zambia, such as manufacturing, where firms similarly depend on reliable and affordable electricity. While adoption pathways may vary with firm size, ownership and access to finance, the importance of credible incentives, clear information flows and coordinated engagement remains consistent. This suggests that the social mechanisms identified in this study are not sector-specific but require contextual adaptation to different operational and regulatory environments.

Beyond organisational and institutional factors, broader acceptance by mining communities and labour groups is also critical to the feasibility and durability of renewable energy projects. Community trust, perceptions of fairness, employment impacts and local participation influence the social licence to operate, particularly where mining activities are closely embedded within surrounding communities. Although these dynamics were not examined directly, they interact closely with organisational decision-making and stakeholder engagement processes.

From a policy perspective, the findings underscore the need for renewable energy governance that prioritises practical implementation over policy expansion. Clearer sector-specific guidance, transparent incentive access, improved information systems and targeted capacity-building would support more consistent adoption. Strengthened partnerships among government, industry, research institutions and communities can further reduce coordination gaps, improve alignment and create more stable conditions for energy investment in the mining sector.

5.7. Research Limitations and Future Research Directions

This study is based on cross-sectional survey data and therefore captures stakeholder perceptions at a single point in time. As a result, the analysis cannot establish causal relationships between social factors and solar energy transition intentions, nor account for how these relationships may evolve as policy frameworks, market conditions and organisational practices change. Future research could address this limitation through longitudinal designs that track transition dynamics over time.

The study also relies on self-reported data collected using a single survey instrument, which introduces the potential for common method bias. While established statistical procedures were applied to assess reliability and validity, future studies could strengthen robustness by combining survey data with objective indicators, documentary analysis or multiple data sources.

In addition, the analysis focuses on perceived policy effectiveness and organisational-level social drivers, without incorporating qualitative interviews or detailed policy text analysis. Further research could complement these findings through in-depth qualitative engagement with policymakers, regulators and industry stakeholders, as well as systematic examination of policy instruments, to better understand institutional contradictions, implementation dynamics and enforcement mechanisms within Zambia's energy governance framework.

The study also does not explicitly examine social acceptance among mining communities and labour groups, which represents an important dimension of sustainable energy transitions. Issues such as community trust, employment impacts and local participation can significantly influence the social licence to operate and warrant focused investigation in future research.

Finally, while the findings are analytically transferable, the study is context-specific to Zambia's mining sector. Comparative research across Sub-Saharan African mining economies with differing regulatory and institutional environments, as well as across other energy-intensive industries such as manufacturing, would help assess the broader applicability of the social mechanisms identified. Such work would provide deeper insight into the conditions that support reliable energy provision and strengthen the operational environment for sustainable energy transitions in industrial sectors.

6. Conclusions

This study examined the social factors shaping solar energy transition in Zambia's mining sector, focusing on policies, incentive systems, communication channels, training

and skills development, and stakeholder engagement and partnerships. The findings indicate that although mining stakeholders recognise persistent energy challenges, progress towards solar adoption remains constrained by weak policy implementation, limited access to incentives, inadequate information flows and insufficient structured training. Among the five factors analysed, stakeholder engagement and incentive systems exerted the strongest influence on transition intentions, while policies, communication and training showed limited statistical significance. These results reinforce existing evidence highlighting the central role of collaboration, regulatory credibility and accessible financial support in advancing renewable energy adoption within energy-intensive industries [5,11,27].

The differentiated effects observed across social factors suggest that effective policy responses must move beyond broad regulatory statements. The strong influence of incentive systems indicates that government interventions should prioritise predictable, sector-specific financial instruments for mining operations, including streamlined tax reliefs, concessional financing and bankable power-purchase arrangements. Similarly, the consistent significance of stakeholder engagement underscores the importance of formalised coordination platforms linking government agencies, mining firms, energy companies, financiers and host communities to reduce uncertainty and improve implementation outcomes.

By contrast, the weak predictive effect of policy frameworks suggests that policy expansion alone is unlikely to accelerate transition. Greater emphasis is required on implementation credibility through enforceable timelines, sector-specific guidance and improved alignment between energy, mining and fiscal institutions. Communication and training initiatives, while necessary, are most effective when integrated with incentive delivery and engagement structures that translate awareness into investment decisions. Overall, the findings indicate that solar energy transition in Zambia's mining sector depends less on policy proliferation and more on targeted incentives, coordinated governance and credible implementation mechanisms aligned with industrial realities.

Building on the empirical results, more targeted and actor-specific actions are required to advance transition. For government institutions, optimising incentive systems involves simplifying access procedures, aligning instruments with the needs of large industrial energy users and providing stable, long-term financial mechanisms that reduce investment uncertainty. Regulatory agencies should strengthen inter-institutional coordination to ensure that policy intentions translate into enforceable, sector-relevant implementation pathways. For mining enterprises, the results highlight the importance of structured stakeholder engagement through partnerships with energy providers, financiers, research institutions and local authorities to share risks, mobilise expertise and support project delivery. Emerging solar pilot initiatives in mining regions illustrate how coordinated collaboration can convert declining solar technology costs into operational energy solutions. At the community level, early and transparent engagement, local participation and effective information sharing remain essential for building trust, addressing labour concerns and maintaining social licence to operate. Collectively, these actions demonstrate how incentive systems and stakeholder participation can be operationalised in practice, consistent with the social factors shown to most strongly influence transition intentions.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/resources15010017/s1>, Table S1: Research Questionnaire; Table S2: Detailed Descriptive Statistics.

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Abbreviations

The following abbreviations are used in this manuscript:

SET	Solar Energy Transition
SE	Sustainable Energy
RE	Renewable Energy
SDG	Sustainable Development Goal
NEP	National Energy Policy
8NDP	Eighth National Development Plans
ADB	African Development Bank
EU	European Union
ERB	Energy Regulation Board
MoE	Ministry of Energy
REFiT	Renewable Energy Feed-in Tariff
RPS	Renewable Portfolio Standard
US	United States
UK	United Kingdom
IPP	Independent Power Producer
PPA	Power Purchase Agreement
NGO	Non-Governmental Organisation
UNDP	United Nations Development Programme
GEF	Global Environment Facility
REEEP	Renewable Energy and Energy Efficiency Partnership
REA	Rural Electrification Authority (Zambia)
ZDA	Zambia Development Agency
GHG	Greenhouse Gas
KMO	Kaiser–Meyer–Olkin
VIF	Variance Inflation Factor
CA	Cronbach’s Alpha
NEC	National Energy Crisis
POLI	Policies
INSYS	Incentive Systems
COMCHA	Communication Channels
TSKLDEV	Training and Skills Development
STKENP	Stakeholder Engagement and Partnerships
STR	Structure system (socio-technical framework)
PPL	People system (socio-technical framework)

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